CM3191 Inorganic Experiment 2: Synthesis and Characterisation of Ferrocene

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Short Lab Report • Thursday laboratory, fume hood B7 • 10 and 17 September 2026
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1. Aims

To prepare ferrocene, $Fe(\eta^5\text{-}C_5H_5)_2$, from cyclopentadiene, KOH and $FeCl_2 \cdot 4\,H_2O$ under nitrogen, purify it by sublimation, and establish its identity, purity and electronic structure by stereomicroscopy, ^{1}H NMR, ATR-IR and UV–Vis spectroscopy, ESI-MS and magnetic susceptibility, each interpreted against the 18-electron, low-spin d^6 sandwich description.[1]

2. Results and Discussion

2.1 Synthesis, yield and purification

Cyclopentadiene (1.0 mL, 0.80 g, 12.10 mmol) was deprotonated by KOH (4.0170 g, 71.6 mmol) in anhydrous DME (12 mL) under nitrogen and treated dropwise with $FeCl_2 \cdot 4\,H_2O$ (1.3274 g, 6.677 mmol) in DMSO (15 mL); quenching on ice with concentrated HCl, filtration, water washes and vacuum drying gave the crude product.[1] The overall reaction,

$$2\,C_5H_6 + 8\,KOH + FeCl_2 \cdot 4\,H_2O \longrightarrow Fe(C_5H_5)_2 + 2\,KCl + 6\,KOH \cdot H_2O, \tag{1}$$

is a deprotonation (pK_a 18 for CpH, driven by the excess of KOH) followed by metathesis of two $C_5H_5^-$ with Fe^{2+}. Cyclopentadiene is limiting (iron in 10% excess), so the theoretical yield is 6.051 mmol × 186.04 g mol^{-1} = 1.126 g (Appendix A). The crude product weighed 1.6127 − 1.5428 = 0.0699 g, a crude yield of only 6.2%. The cyclopentadienide solution was only "pinkish brown" after 10 min rather than deep burgundy, which is consistent with incomplete deprotonation when the iron salt was added, and the green colour on quenching is consistent with unreacted Fe^{2+} (Appendix D).

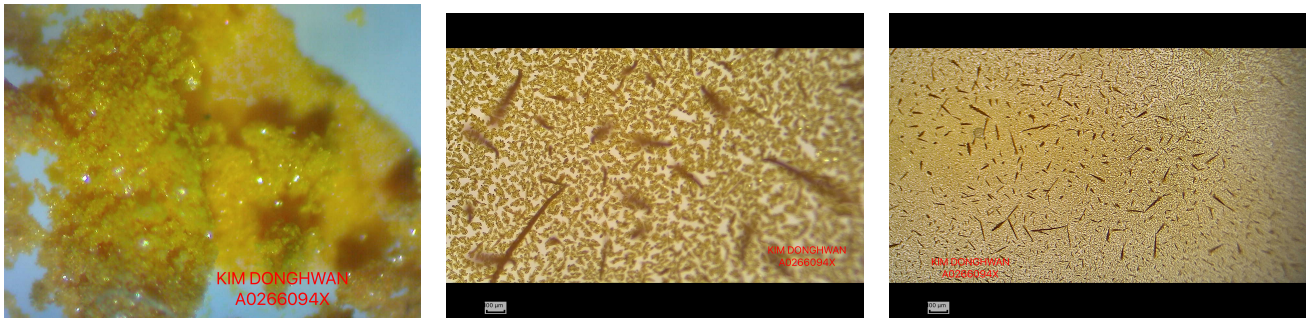
Sublimation at 100–140 °C with an ice-cooled cover gave 0.0403 + 0.0123 = 0.0526 g of orange crystalline ferrocene from the original crude, a recovery of 75% and an isolated yield of 4.7% on the nominal cyclopentadiene charge. After the microscopy, NMR, ESI-MS and IR aliquots only 9 mg remained, less than the 20–25 mg needed for UV–Vis and the 0.10–0.12 g for the magnetic balance, so a second portion of crude (0.3021 g, weighed on tared paper; Appendix E) was issued and sublimed: 0.1104 + 0.0563 + 0.0138 = 0.1805 g (59.7% recovery). The UV–Vis and magnetic samples came from this second sublimate (Table 1); the yield refers to the original synthesis only. The two recoveries, 75% and 60%, show that sublimation was efficient and that the low isolated yield was set almost entirely by the crude yield, i.e. by the reaction and work-up; the losses in sublimation (water driven off in cycle 1, product lost each time the cover is lifted, residue on the base) were secondary. Because the sublimate is spectroscopically pure ferrocene (below), the 75% recovery also shows that at least three-quarters of the crude mass was ferrocene: the crude was not grossly contaminated with water or salt.

Table 1: Masses, yields and sample provenance (analytical balance, ±0.0001 g; Appendix D).

Quantity	Mass / g	Yield or recovery	Used for
Theoretical ferrocene (CpH limiting)	1.1258	—	
Crude ferrocene (1.6127 − 1.5428)	0.0699	6.2%	sublimation, microscopy
Sublimate from the original crude (cycles 1, 2)	0.0526	75.3% rec.; 4.7% overall	microscopy, NMR, ESI-MS, IR
Second crude portion	0.3021	—	
Sublimate from the second portion (cycles 3, 4, dish)	0.1805	59.7% rec.	UV–Vis (0.0232 g), MSB (0.0867 g)

Under the stereomicroscope the crude is a compact mass of yellow–orange granules without defined faces, as expected for a rapidly precipitated powder, whereas both sublimates consist of discrete, flat, angular orange plates and needles with sharp edges, the habit of ferrocene grown slowly from the vapour

(Figure 1b,c); the change of habit is the qualitative evidence of purification, and the two batches are indistinguishable by eye.



(a) crude, Session I

(b) sublimed, original batch (100 μm bar)

(c) sublimed, second portion (100 μm bar)

Figure 1: Stereomicrographs at the manual's magnification of (a) the crude ferrocene, (b) the sublimate of the original crude (used for NMR, ESI-MS and IR) and (c) the sublimate of the second crude portion (used for UV–Vis and the magnetic balance); original files with embedded scale bars. A higher-magnification field of (c) is shown in Figure 4.

2.2 ^{1}H NMR spectroscopy

The spectrum of the sublimate (ca. 1 mg in 0.6 mL $CDCl_3$; 400.23 MHz, 297.9 K) is shown in full with every signal integrated in Figure 2, with the data-set name `CM3191-Thurs-400-Sep17-2026-09` on the plot; the axis is referenced to residual $CHCl_3$ at 7.26 ppm.[2] Signals, integrals relative to the ring singlet: δ 7.26 (s, 1.71, $CHCl_3$); **4.17 (s, 10.00, Fe(C_5H_5)$_2$)**; 2.17 (s, 0.21, acetone); 1.55 (s, 3.73, H_2O); 1.25 (br s, 0.13, aliphatic); literature 4.15 (s).[3]

A single sharp line (FWHM 2.0 Hz) for all ten ring protons is what the sandwich structure requires: the two η^5-rings are equivalent and rotate freely, so all C–H are identical on the NMR time scale. The weak lines at 4.389 and 3.953 ppm are the ^{13}C satellites of the singlet, not impurities: they are symmetric about it, separated by $^{1}J_{CH} = 175$ Hz, and their combined integral (1.2%) matches the 1.1% expected for one ^{13}C among ten CH units (Figure 5). No olefinic (5–7 ppm, dienes), DME (3.40, 3.55) or DMSO (2.62) signal appears above the noise. From the noise of the empty 9–10 ppm region, a sharp singlet would be detected at an integral of about 0.01 on this scale, which corresponds to roughly 1 mol% of a one-proton impurity or 0.1–0.2 mol% of the six- to twelve-proton candidates above (an estimate; no quantitative standard was run). The water (1.9 mol per mol ferrocene, about 0.2 mg in the tube) and acetone most likely entered with the solvent and the acetone-rinsed glassware; their origin cannot be assigned to the solid without a weighed sample and a solvent blank.

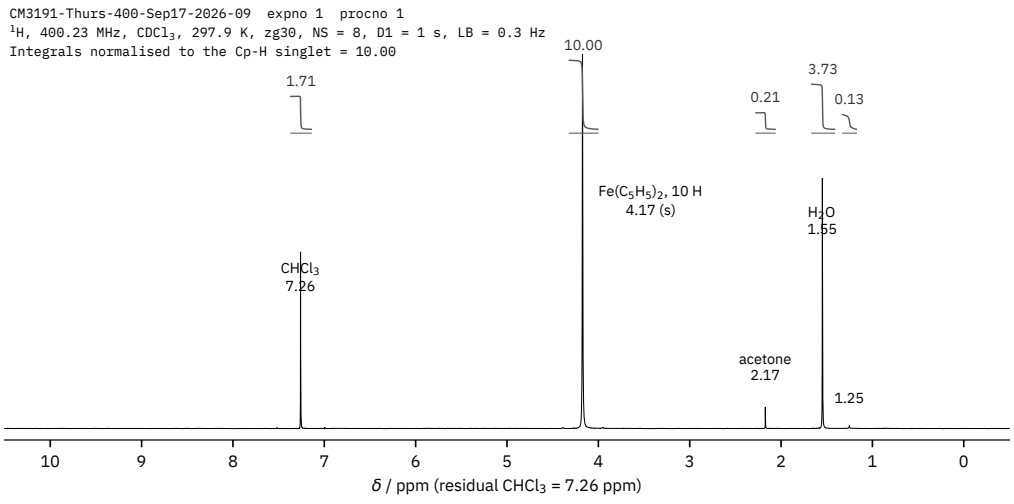


Figure 2: Full 0–10 ppm ^{1}H NMR spectrum of the sublimed ferrocene with all signals integrated (Cp-H = 10.00) and the data-set name shown; 400.23 MHz, $CDCl_3$, 297.9 K, zg30, 8 scans, D1 = 1 s, LB = 0.3 Hz; referenced to $CHCl_3$ = 7.26 ppm. Expansions in Figure 5.

2.3 ATR-IR spectroscopy

The solid-state spectrum (500–4000 cm^{-1}, Figure 6) shows the five infrared-active fundamentals of ferrocene, each within 10 cm^{-1} of the Lippincott–Nelson assignments:[4] 3095 (lit. 3085, ring C–H stretch; no aliphatic C–H below 3000), 1407 (1411, antisymmetric C–C stretch), 1104 (1108, ring breathing), 999 (1002, in-plane C–H bend) and 812 cm^{-1} (811, out-of-plane C–H bend, the strongest band). No O–H absorption above 3200 cm^{-1} and no C=O or C=C band near 1600 cm^{-1} is detectable, so neither water nor dienes nor oxidation products are present at the level ATR-IR can see; the weak features at 1047, 855 and 786 cm^{-1} lie where ferrocene shows combination bands.

2.4 UV–Vis spectroscopy and ligand-field assignment

Ferrocene (0.0232 g) in methanol (25.00 mL), $c = 0.0232/(186.04 \times 0.02500) = 4.988 \times 10^{-3}$ M, was measured undiluted in a 1.00 cm cell, both maxima falling inside the 0.1–1.0 absorbance window ($D = 1$): 440.50 nm ($A = 0.416$) and 325.00 nm ($A = 0.239$) (Figure 7). From $A = \varepsilon c l$,

$$\varepsilon_{440} = \frac{0.416}{4.988 \times 10^{-3} \times 1.00} = 83\ \mathrm{M^{-1}\,cm^{-1}}, \qquad \varepsilon_{325} = \frac{0.239}{4.988 \times 10^{-3} \times 1.00} = 48\ \mathrm{M^{-1}\,cm^{-1}}. \tag{2}$$

Sohn, Hendrickson and Gray report 22 700 cm^{-1} (440.5 nm, ε 91) and 30 800 cm^{-1} (324.7 nm, ε 49) in EPA: the positions agree within 1 nm, the absorptivities to 9% and 2%.[5,6] Absorptivities below 100 mark both bands as Laporte-forbidden d–d transitions. For the d^6 metallocene (ground state $^1A_{1g}$, $(e_{2g})^4(a_{1g})^2$) the 440 nm envelope contains the two spin-allowed transitions $^1A_{1g} \rightarrow a\,^1E_{1g}$ and $^1A_{1g} \rightarrow {}^1E_{2g}$, and the 325 nm band is $^1A_{1g} \rightarrow b\,^1E_{1g}$; all three move an electron from the filled, essentially metal e_{2g}/a_{1g} orbitals into the empty antibonding e_{1g}^* level.[5] The 9% shortfall at 440 nm exceeds the uncertainty propagated from typical tolerances (about 1%, Appendix A) and is more plausibly due to the solvent, a baseline offset or non-absorbing mass in the weighed solid.

2.5 ESI-MS

The positive-ion electrospray spectrum of the sublimate (sample SN9; Thermo LTQ XL ion trap, flow injection in methanol, capillary 43 V, no added electrolyte; m/z 70–1400; Figure 3) has one dominant ion at m/z 185.96 with satellites at 184.04 (6.5%), 186.96 (14.2%) and 188.04 (1.2%); nothing else exceeds 5%.[10] Ferrocene has neither a basic nor a coordinating site, so it forms no $[M + H]^+$ and, despite the sodium present in the system, no $[M + Na]^+$ (209, absent); because its oxidation potential is low it is instead oxidised at the emitter, the electrospray source acting as an electrolytic cell, and is detected as the ferrocenium radical cation $[Fe(C_5H_5)_2]^{+\bullet}$.[9] The envelope calculated for $C_{10}H_{10}Fe^+$, m/z 184 (6.4%, ^{54}Fe), 185 (0.7), 186 (100), 187 (13.2), 188 (1.1), is reproduced peak for peak (Appendix B).[7] The 184 peak, absent for any purely organic ion, is the signature of one iron atom, and the 187 peak is accounted for by the ^{13}C satellite (10.7% from ten carbons plus 2.1% from ^{57}Fe), leaving little room for $[M+H]^+$. No dimer ion (373), Cp-loss fragment (121) or substituted-ferrocene ion is detected. The spectrum confirms the molecular formula; because ESI intensities are not quantitative for composition, it does not by itself give a bulk purity.

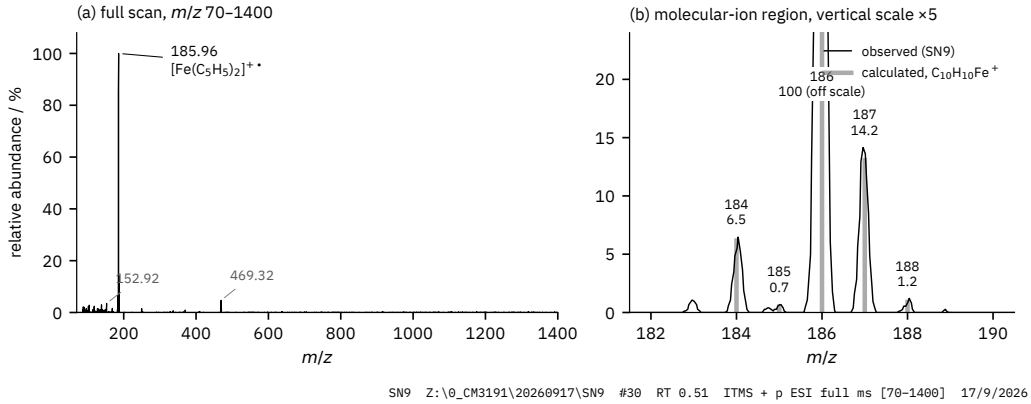


Figure 3: Positive-ion ESI mass spectrum of the sublimed ferrocene (SN9): (a) full scan; (b) molecular-ion region with the envelope calculated for $C_{10}H_{10}Fe^+$ (grey bars). Re-plotted from the instrument export; original printout in Appendix C.

2.6 Magnetic susceptibility

The ground sublimate was packed to 17 mm in the tube of a Sherwood MSB AUTO (unit 1); $m = 0.9090 - 0.8223 = 0.0867$ g (Appendix A). The displays read $\chi_v = -0.005 \times 10^{-4}$ and $\chi_g = 0.000 \times 10^{-4}$ $cm^3\ g^{-1}$; the

instrument-room temperature recorded by the laboratory staff was 23 °C (296 K).[10] The five photographs required by the procedure are Figure 8.

The literature susceptibility of ferrocene, $\chi_M = -125 \times 10^{-6}$ cm^3 mol^{-1},[6] corresponds to $\chi_g = -6.7 \times 10^{-7}$ cm^3 g^{-1}, which this display would show as -0.007×10^{-4}. The instrument showed 0.000 in mass mode and a small negative value in volume mode: the expected diamagnetism is of the same size as the balance's zero offset (no repeat zero or blank was recorded), so it is not resolved numerically, and the negative volume-mode reading is only qualitatively consistent with it. What the measurement does establish is the absence of unpaired electrons. One unpaired electron ($\mu_{\text{eff}} = 1.73\ \mu_B$) would give $\chi_M = \mu^2/8T \approx 1.3 \times 10^{-3}$ cm^3 mol^{-1} at 296 K, i.e. $\chi_g = +6.8 \times 10^{-6}$ cm^3 g^{-1} or a display of $+0.068 \times 10^{-4}$, roughly ten times the zero offset. Nothing of the kind was seen, so μ_{eff} is zero within the resolution of the balance; no numerical moment is assigned. This is the expected result: the six d electrons fill e_{2g} and a_{1g} below the e^*_{1g} gap, and the 18-electron count (d^6 plus two six-electron Cp$^-$ ligands) is closed-shell and low-spin, which is also why the NMR line is sharp and unshifted. Applying the Pascal correction ($+125 \times 10^{-6}$) to a displayed zero would manufacture a spurious μ_{eff} of about 0.55 μ_B; the correction belongs to a resolved measured value.[8] The packed mass was slightly below the manual's 0.10–0.12 g.

2.7 Sources of error

Yield losses arise from incomplete deprotonation, re-dimerisation and evaporation of cyclopentadiene, air oxidation of Fe^{2+} during the slow addition, under-delivery of the 1.0 mL charge, product left in flask, funnel and filter, and losses on lifting the sublimation cover; any water or KCl in the crude inflates the crude yield and deflates the apparent recovery, but the 75% recovery bounds such contamination to under a quarter of the crude mass. The molar absorptivities carry about 1% uncertainty from typical balance, flask and photometric tolerances (Appendix A), so the 9% difference at 440 nm is not random; a non-absorbing impurity alone cannot explain it (it would have to be 8% at 440 nm but 2% at 325 nm), so solvent and baseline effects contribute. The magnetic balance's zero offset exceeds its last displayed digit, so it excludes paramagnetism rather than measuring χ.

3. Conclusion

Ferrocene was synthesised from cyclopentadiene and $FeCl_2 \cdot 4\,H_2O$ via potassium cyclopentadienide and purified by sublimation: 4.7% isolated yield on the nominal cyclopentadiene charge (6.2% crude; sublimation recoveries 75% and 60%), so the loss occurred in the reaction and work-up, not in the purification. Every technique identifies the sublimate as ferrocene: one sharp ^{1}H resonance at δ 4.17 for ten equivalent protons with its ^{13}C satellites and no other synthesis-derived signal above about 1 mol% per proton; the five infrared fundamentals at 3095, 1407, 1104, 999 and 812 cm^{-1}; ligand-field bands at 440.5 nm ($\varepsilon = 83$) and 325.0 nm ($\varepsilon = 48$) within 1 nm of the literature; and a ferrocenium ion at m/z 186 whose ^{54}Fe and ^{13}C satellites match the calculated envelope. The magnetic balance showed no paramagnetism at a level ten times above its zero offset: the complex is a closed-shell, low-spin, 18-electron d^6 sandwich with no unpaired electrons.

References

[1] Chan, E. *CM3191 Chemical Synthesis Experiments: Lab Manuals for Inorganic Experiments*, version 0626; National University of Singapore, AY2026/27; Experiment 2, pp 12–24.

[2] Fulmer, G. R.; Miller, A. J. M.; Sherden, N. H.; Gottlieb, H. E.; Nudelman, A.; Stoltz, B. M.; Bercaw, J. E.; Goldberg, K. I. NMR Chemical Shifts of Trace Impurities: Common Laboratory Solvents, Organics, and Gases in Deuterated Solvents Relevant to the Organometallic Chemist. *Organometallics* **2010**, *29*, 2176–2179.

[3] Powers, T. M. Structure of Ferrocene. *JoVE Science Education: Inorganic Chemistry*; JoVE: Cambridge, MA, 2017. `https://app.jove.com/v/10347` (accessed 8 October 2026).

[4] Lippincott, E. R.; Nelson, R. D. The Vibrational Spectra and Structure of Ferrocene and Ruthenocene. *Spectrochim. Acta* **1958**, *10*, 307–329.

[5] Sohn, Y. S.; Hendrickson, D. N.; Gray, H. B. Electronic Structure of Metallocenes. *J. Am. Chem. Soc.* **1971**, *93*, 3603–3612.

[6] Sohn, Y. S. *Electronic Structures of Metallocene Complexes*. Ph.D. Dissertation, California Institute of Technology, 1971; Part I, Table II; Part II, Table I (χ_M after Engelmann, *Z. Naturforsch.* **1953**, *8b*, 775). DOI: 10.7907/pg0q-y417.

[7] Coursey, J. S.; Schwab, D. J.; Tsai, J. J.; Dragoset, R. A. *Atomic Weights and Isotopic Compositions*, version 4.1; National Institute of Standards and Technology: Gaithersburg, MD. `https://physics.nist.gov/Comp` (accessed 7 October 2026).

[8] Bain, G. A.; Berry, J. F. Diamagnetic Corrections and Pascal's Constants. *J. Chem. Educ.* **2008**, *85*, 532–536.

[9] Van Berkel, G. J.; Zhou, F. Characterization of an Electrospray Ion Source as a Controlled-Current Electrolytic Cell. *Anal. Chem.* **1995**, *67*, 2916–2923. DOI: 10.1021/ac00113a028.

[10] Chan, E. CM3191 Inorganic Canvas announcements, 24 September 2026 (ESI-MS instrumental settings) and 26 September 2026 (temperature in the magnetic measurement). National University of Singapore.

Figures

Figure 4: Higher-magnification stereomicrograph of the sublimate of the second crude portion (60 μm bar; original file with embedded label). Supplementary to Figure 1c, which is at the magnification prescribed by the manual.

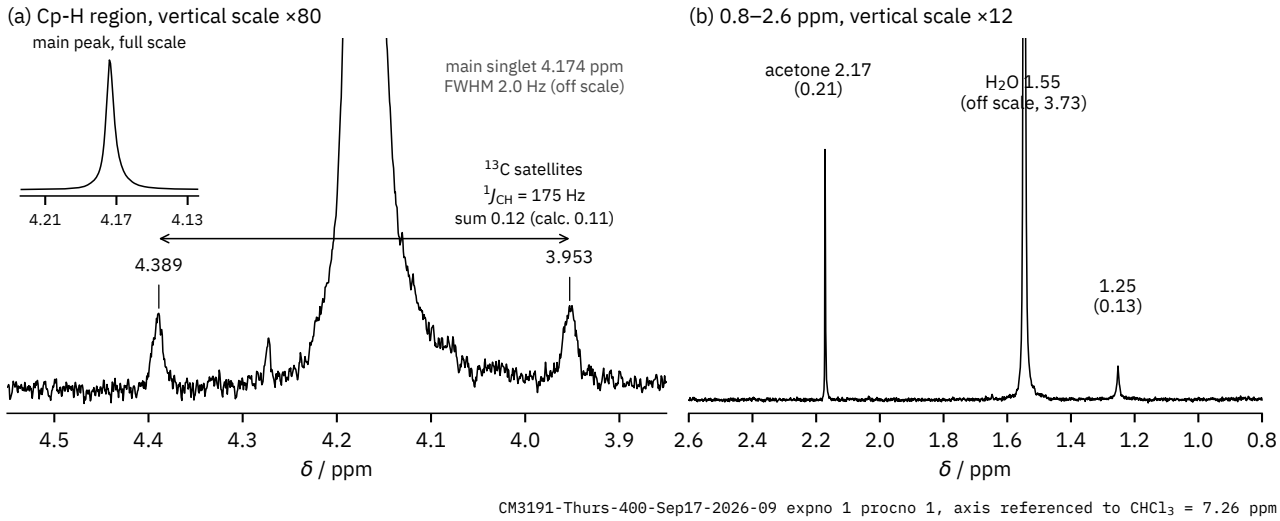


Figure 5: Expansions of Figure 2. (a) Cp-H region at 80-fold vertical gain showing the ^{13}C satellites ($^{1}J_{CH}$ = 175 Hz) and, inset, the parent singlet at full scale; (b) 0.8–2.6 ppm at 12-fold gain (integrals in brackets, Cp-H = 10.00).

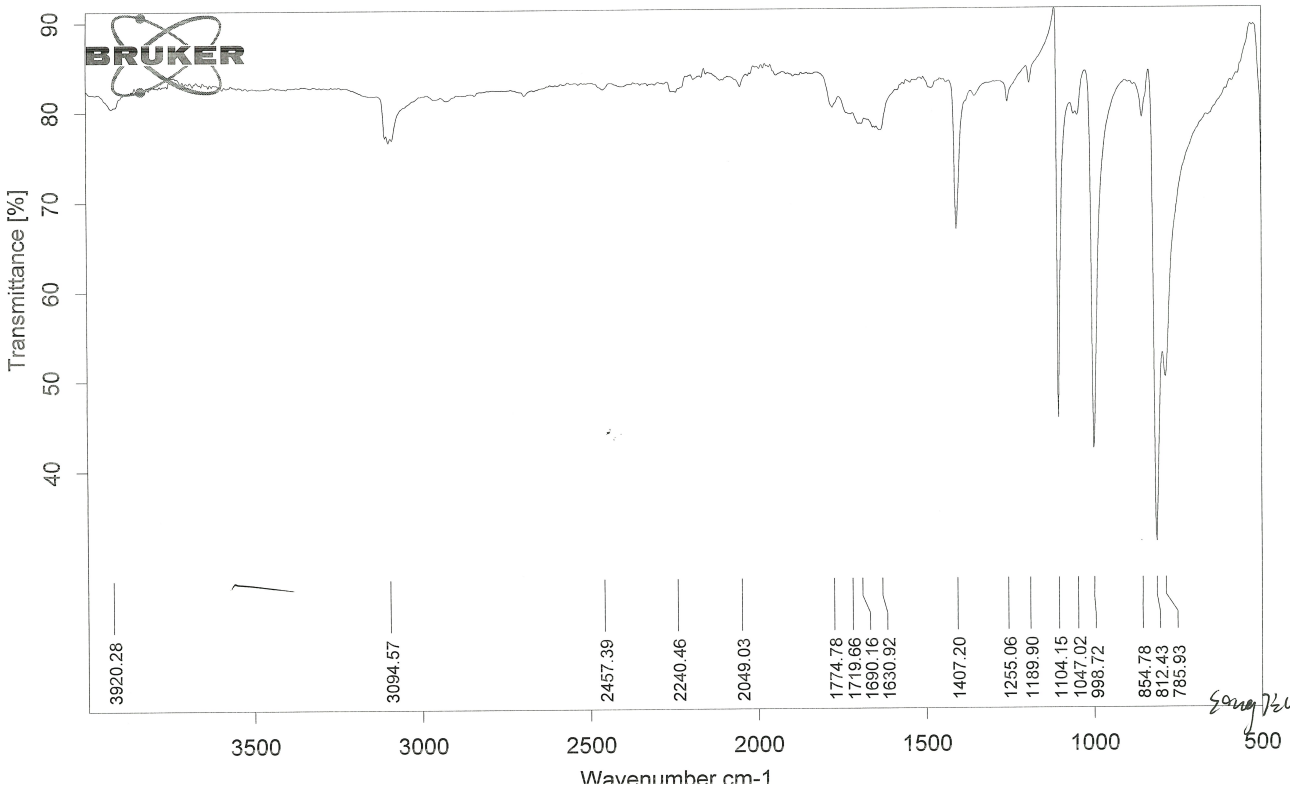


Figure 6: ATR-IR spectrum of the sublimed ferrocene, 500–4000 cm^{-1}, 17 September 2026 (detail of the signed printout; complete page in Appendix C).

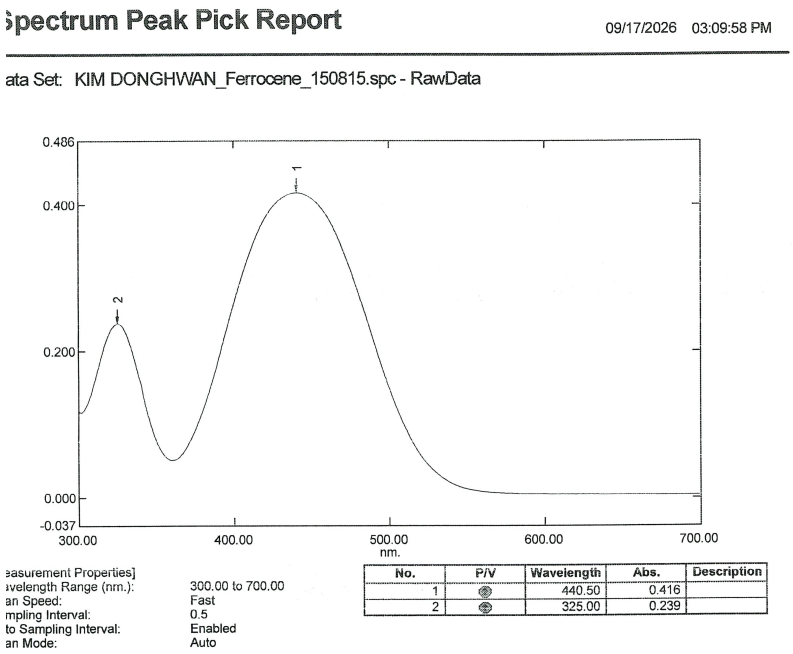
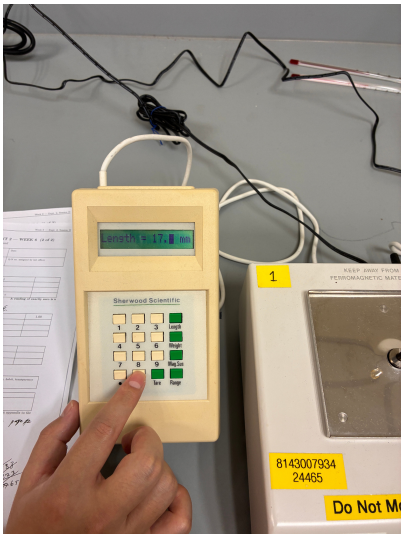
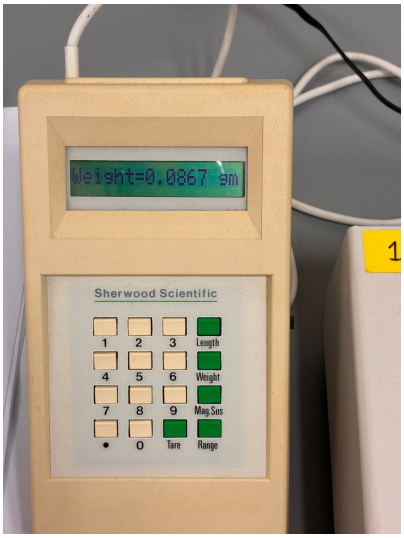


Figure 7: UV–Vis spectrum (300–700 nm, methanol, 1.00 cm) of the sublimed ferrocene, 4.988×10^{-3} M, with the instrument peak-pick table (440.50 nm, $A = 0.416$; 325.00 nm, $A = 0.239$); the signed printout is in Appendix C. CONFIRM: handwrite the ε calculation on the printout and replace the scan

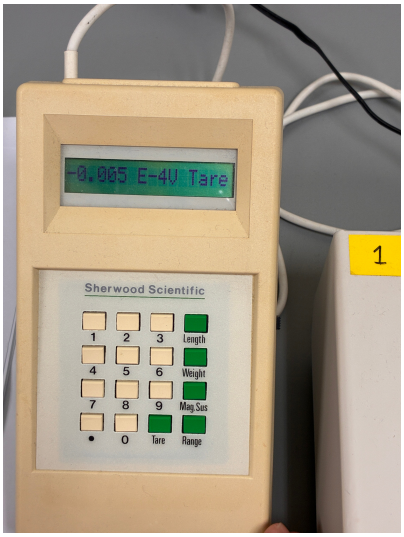
(1) zero reading before the tube was placed (`IMG_8576`; display washed out by reflection)



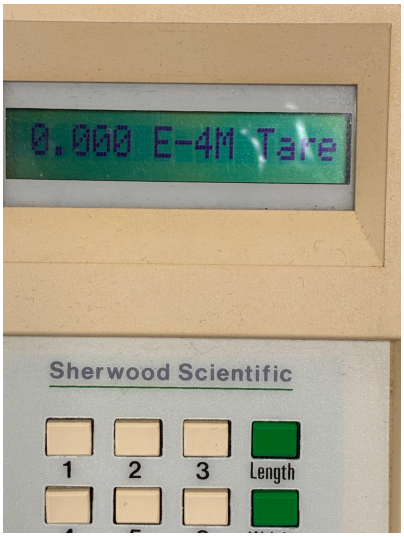
(2) length entry before the last digit, `17._ mm` (`IMG_8574`; detector labels unit 1 / 8143007934)



(3) weight entry, `Weight=0.0867 gm` (`IMG_8575`)



(4) volume susceptibility, `-0.005 E-4V` (`IMG_8577`)



(5) mass susceptibility, `0.000 E-4M` (`IMG_8579`)

Figure 8: The five photographs required by the manual for the Sherwood MSB AUTO measurement (unit 1), taken during the measurement under the demonstrator's supervision, in the order of the procedure: zero reading, length entry, weight entry, χ_v and χ_g results. Original, uncropped files. The weighing and packed-length photographs and enlarged display crops are in Appendix E.

Appendix A. Calculations

Limiting reagent and theoretical yield.

$$n(\mathrm{C_5H_6}) = \frac{1.0\ \mathrm{mL} \times 0.80\ \mathrm{g\,mL^{-1}}}{66.10\ \mathrm{g\,mol^{-1}}} = 12.10\ \mathrm{mmol}, \qquad n(\mathrm{FeCl_2 \cdot 4\,H_2O}) = \frac{1.3274}{198.81} = 6.677\ \mathrm{mmol}, \tag{3}$$

$$n(\mathrm{KOH}) = \frac{4.0170}{56.11} = 71.6\ \mathrm{mmol\ (8\ equiv = 48.4\ mmol)}, \qquad n(\mathrm{Fc})_{\mathrm{theor}} = \tfrac{1}{2}\, n(\mathrm{C_5H_6}) = 6.051\ \mathrm{mmol}, \tag{4}$$

$$m_{\mathrm{theor}} = 6.051 \times 10^{-3} \times 186.04 = 1.1258\ \mathrm{g} \quad \text{(CpH limiting; Fe in 10\% excess)}. \tag{5}$$

Yields and recoveries.

$$m_{\mathrm{crude}} = 1.6127 - 1.5428 = 0.0699\ \mathrm{g}, \qquad Y_{\mathrm{crude}} = 0.0699/1.1258 = 6.21\%, \tag{6}$$

$$m_{\mathrm{subl,orig}} = 0.0403 + 0.0123 = 0.0526\ \mathrm{g}, \qquad Y_{\mathrm{isolated}} = 0.0526/1.1258 = 4.67\%, \tag{7}$$

$$R_{\mathrm{subl,orig}} = 0.0526/0.0699 = 75.3\%, \qquad m_{\mathrm{residue}} = 1.1083 - 1.0989 = 0.0094\ \mathrm{g}, \tag{8}$$

$$m_{\mathrm{subl,2nd}} = 0.1104 + 0.0563 + 0.0138 = 0.1805\ \mathrm{g}, \qquad R_{\mathrm{subl,2nd}} = 0.1805/0.3021 = 59.7\%. \tag{9}$$

Molar absorptivity and uncertainty (undiluted stock, $D = 1$; tolerances are typical values, not measured repeatabilities).

$$c = \frac{0.0232\ \mathrm{g}}{186.04\ \mathrm{g\,mol^{-1}} \times 0.02500\ \mathrm{L}} = 4.988 \times 10^{-3}\ \mathrm{M}, \tag{10}$$

$$\varepsilon = \frac{A}{cl}\ (D = 1)\ : \qquad \varepsilon_{440} = 83.4, \qquad \varepsilon_{325} = 47.9\ \mathrm{M^{-1}\,cm^{-1}}, \tag{11}$$

$$\frac{u(\varepsilon)}{\varepsilon} = \sqrt{\left(\frac{u(A)}{A}\right)^2 + \left(\frac{u(m)}{m}\right)^2 + \left(\frac{u(V)}{V}\right)^2 + \left(\frac{u(l)}{l}\right)^2} = 0.7\%\ (440\ \mathrm{nm}),\ 1.0\%\ (325\ \mathrm{nm}), \tag{12}$$

with $u(A) = 0.002$, $u(m) = 0.0001$ g, $u(V) = 0.03$ mL, $u(l) = 0.002$ cm.

Magnetic susceptibility. The empty-tube mass 0.8223 g appears in both handwritten subtractions on the datasheet ($0.8988 - 0.8223 = 0.0765$, crossed out; $0.9090 - 0.8223 = 0.0867$, entered); the "0.8264" written in the second is a copying slip, since only 0.8223 reproduces the entered mass. The comparison values below are what the display would show for the literature diamagnetism and for one unpaired electron; no experimental moment is derived, because the zero offset of the balance was not characterised.

$$m = 0.9090 - 0.8223 = 0.0867\ \mathrm{g}; \quad \text{displayed } \chi_v = -0.005 \times 10^{-4},\ \chi_g = 0.000 \times 10^{-4}\ \mathrm{cm^3\,g^{-1}}, \tag{13}$$

$$\chi_{g,\mathrm{lit}} = \frac{-125 \times 10^{-6}}{186.04} = -6.7 \times 10^{-7}\ \mathrm{cm^3\,g^{-1}}\ (\mathrm{display}\ -0.007 \times 10^{-4}), \tag{14}$$

$$\chi_g(S = \tfrac{1}{2}) = \frac{\mu^2}{8TM} = \frac{1.73^2}{8 \times 296.15 \times 186.04} = +6.8 \times 10^{-6}\ \mathrm{cm^3\,g^{-1}}\ (\mathrm{display}\ +0.068 \times 10^{-4}), \quad T = 23\ ^\circ\mathrm{C}. \tag{15}$$

Appendix B. Calculated ESI isotope envelope for $\mathrm{C_{10}H_{10}Fe^+}$

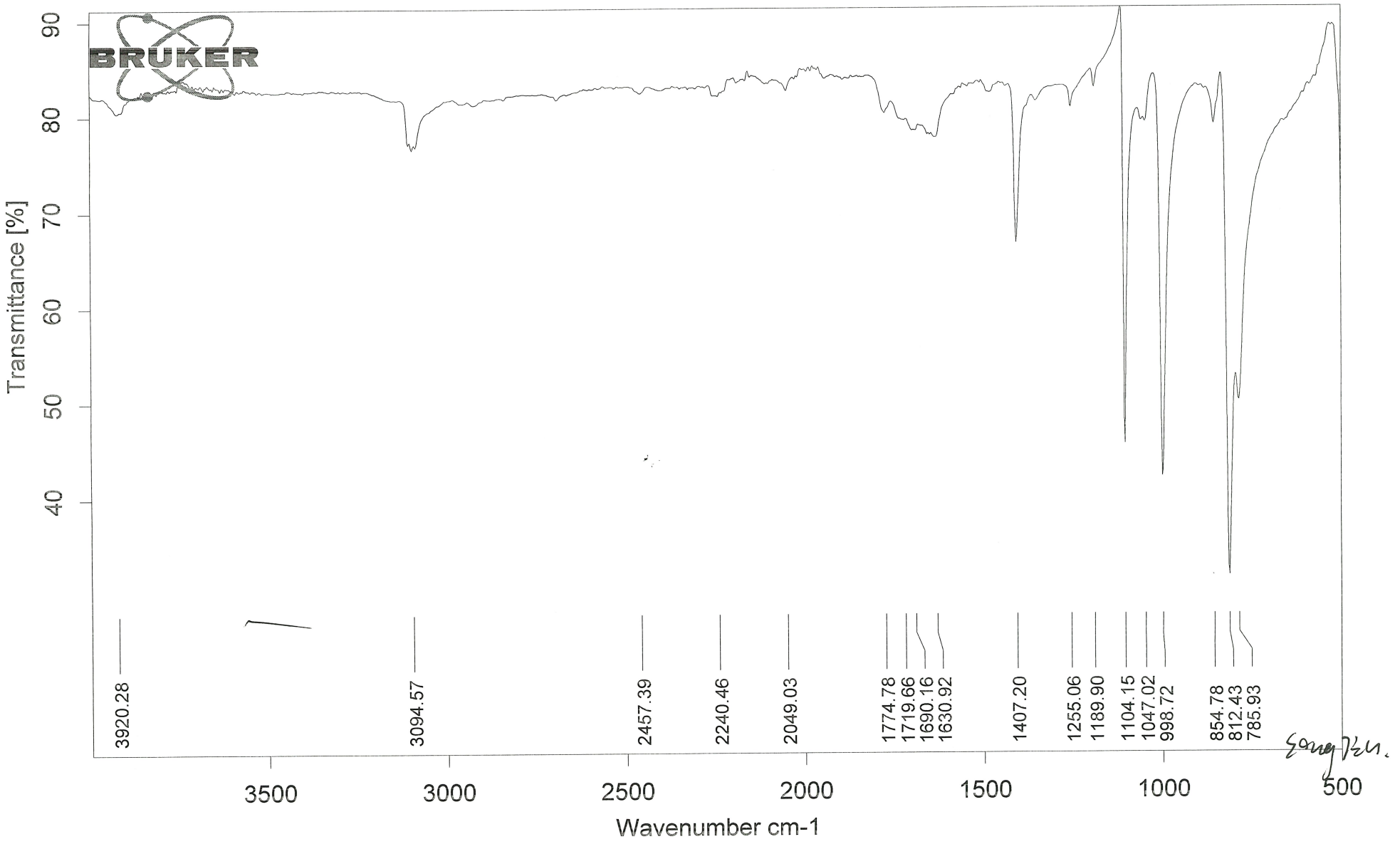
Natural-abundance distributions[7] convolved for ten C, ten H and one Fe; the ^{56}Fe group at nominal m/z = 186 is normalised to 100 (singly charged ion, no adducts). Observed values were read from the vector trace of the instrument printout (base peak = 100); m/z labels are the instrument's.

Nominal m/z	Calculated / %	Observed / %	Origin
184	6.37	6.5	^{54}Fe
185	0.70	0.7	^{54}Fe + ^{13}C
186	100.0	100	$^{12}C_{10}\,^{1}H_{10}\,^{56}$Fe
187	13.24	14.2	^{13}C (10.7%) or ^{57}Fe (2.1%)
188	1.10	1.2	$^{13}C_2$, ^{58}Fe, or ^{13}C + ^{57}Fe
189	0.06	—	

First-order check: $I_{M+1}/I_M \approx 10\,x(^{13}\mathrm{C})/x(^{12}\mathrm{C}) + x(^{57}\mathrm{Fe})/x(^{56}\mathrm{Fe}) + 10\,x(^{2}\mathrm{H})/x(^{1}\mathrm{H}) \approx 0.108 + 0.023 + 0.001 = 0.132$.

Appendix C. Original IR, UV–Vis and ESI-MS printouts

The IR and UV–Vis pages are the signed originals. The ESI-MS pages are the instrument export of 17 September 2026 (file `SN9-KIM_DONGHWAN.docx`), rendered without alteration.

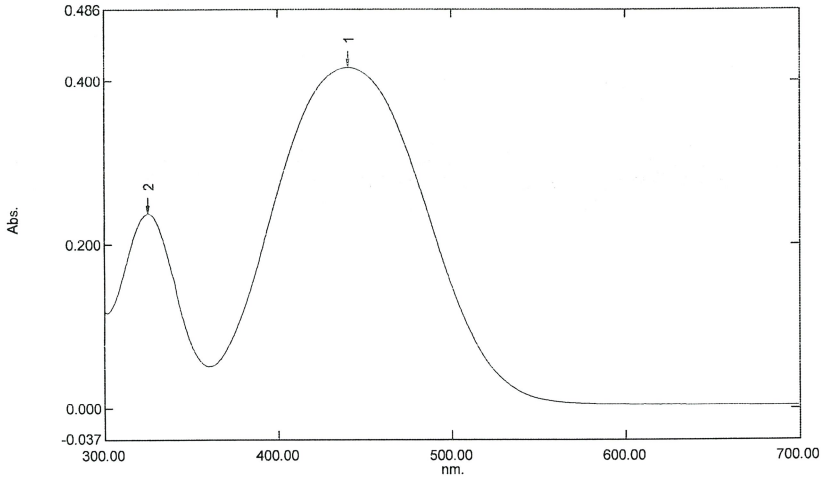


D:\CM3191\17Sep2026\Ferrocene_Kim_donghwan.0	Ferrocene_Kim_donghwan	Solid	17/9/2026

Spectrum Peak Pick Report

09/17/2026 03:09:58 PM

Data Set: KIM DONGHWAN_Ferrocene_150815.spc - RawData



No.	P/V	Wavelength	Abs.	Description
1		440.50	0.416	
2		325.00	0.239	

[Measurement Properties]
Wavelength Range (nm.): 300.00 to 700.00
Scan Speed: Fast
Sampling Interval: 0.5
Auto Sampling Interval: Enabled
Scan Mode: Auto

[Instrument Properties]
Instrument Type: UV-1800 Series
Measuring Mode: Absorbance
Slit Width: 1.0 nm
Light Source Change Wavelength: 340.0 nm
S/R Exchange: Normal

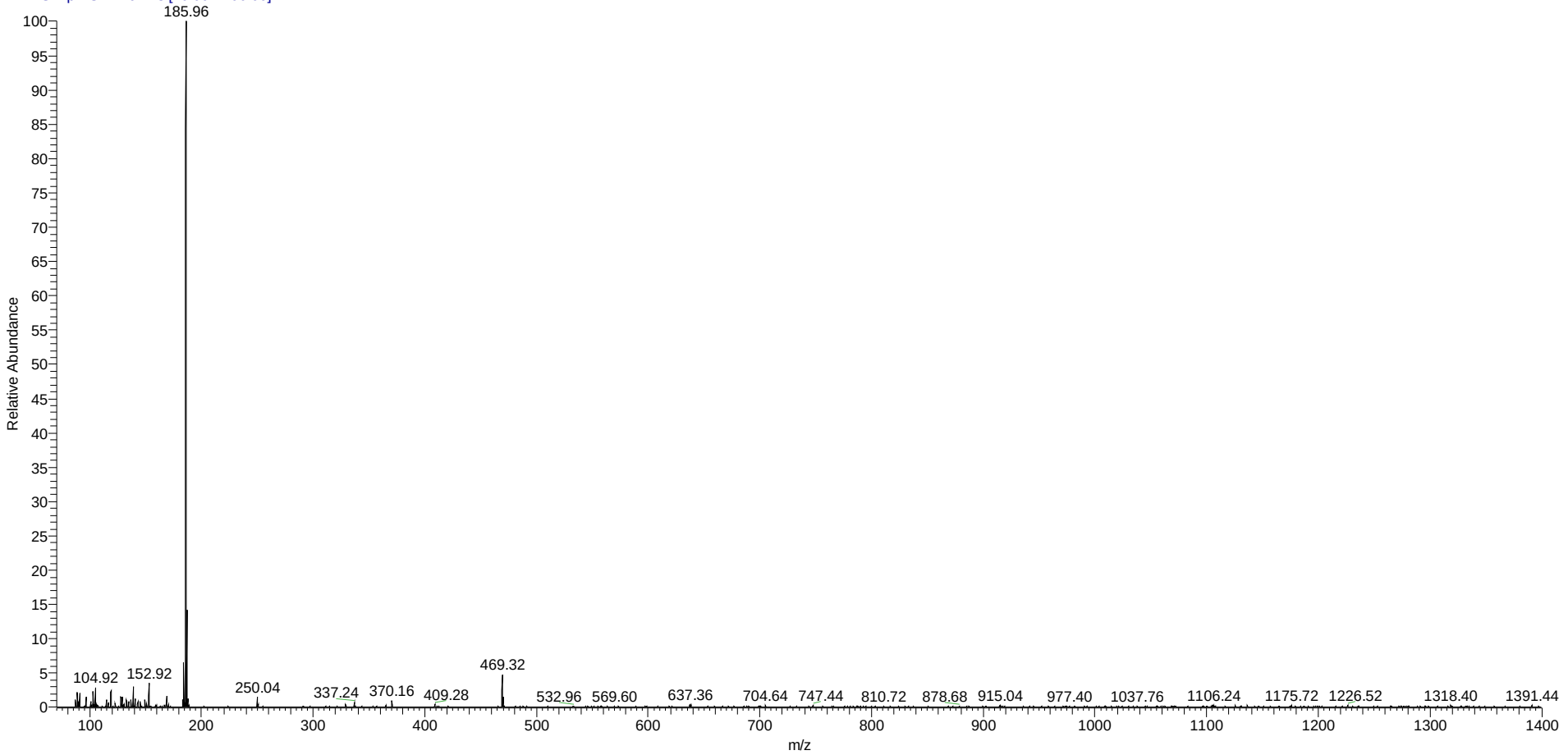
[Attachment Properties]
Attachment: None

[Operation]
Threshold: 0.0010000
Points: 4
InterPolate: Disabled
Average: Disabled

[Sample Preparation Properties]
Weight:
Volume:
Dilution:
Path Length:
Additional Information:

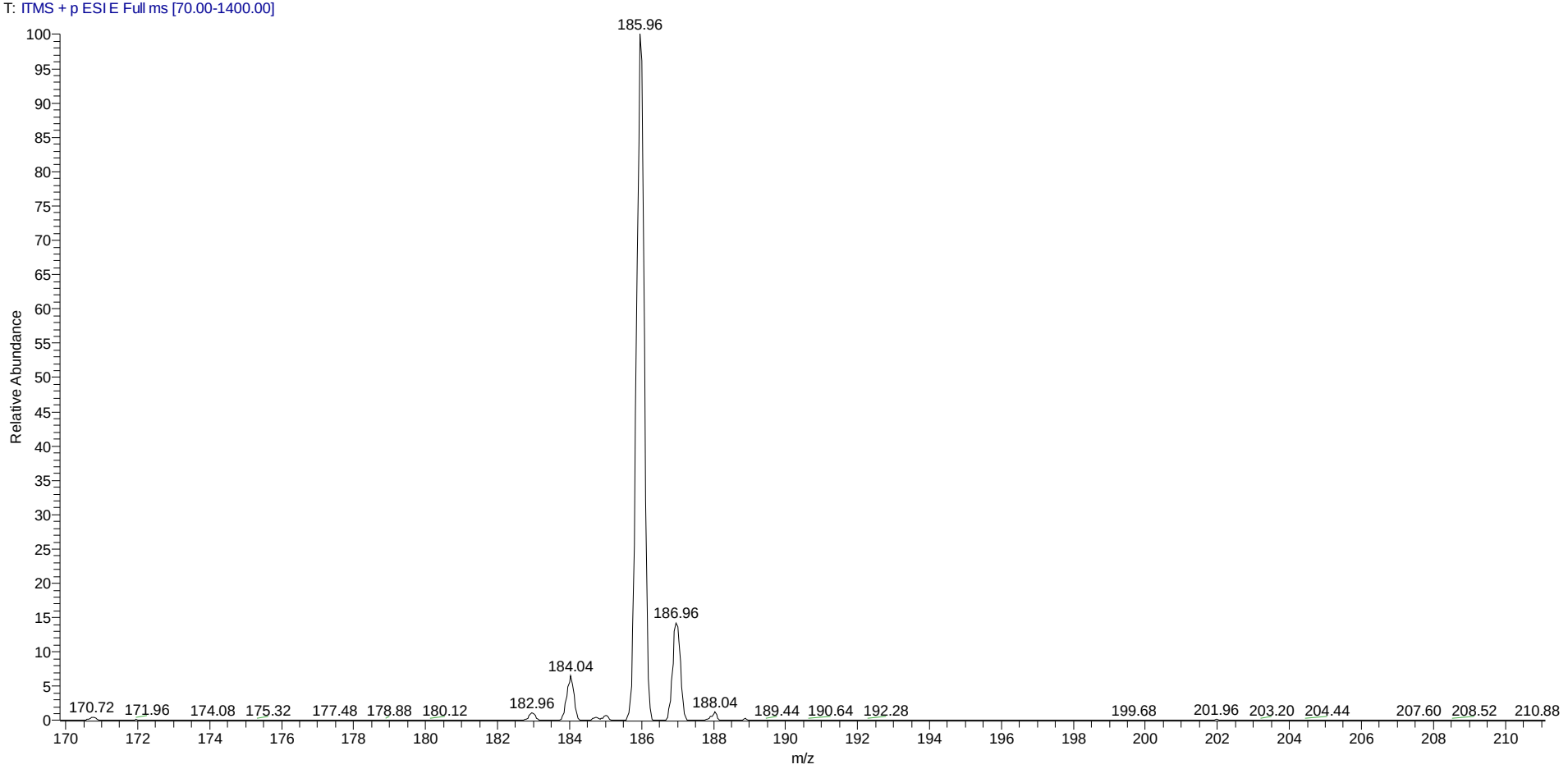
Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Z:\0_CM3191\20260917\SN9 17/9/2026 5:56:29 pm Enter Module Name
Enter Student Name

SN9 #30 RT: 0.51 AV: 1 NL: 1.41E5
T: ITMS + p ESI E Full ms [70.00-1400.00]



Appendix D. Signed datasheets

Reproduced without alteration (`TalkFile_Week5.pdf` p. 4; `TalkFile_Week6.pdf` pp. 2, 4, 5).

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 5

Session I: Synthesis and Microscopic Image of Crude Ferrocene

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

~30.03 g ice

Reagent	Nominal	Actual	Remarks
KOH pellets / anhydrous DME	4.0 g / 12 mL	4.0170 g	
Cyclopentadiene monomer	1.0 mL		
$FeCl_2 \cdot 4H_2O$ / DMSO (dissolve, rinse)	1.3 g / 7, 3.5 mL	1.3274 g	
Conc. HCl / ice	10 mL / 30 g		
DI water (rinse / wash 3×)	10 / 20 mL		

2. Conditions

	Nominal	Actual	Remarks
N_2 purge before CpH addition	3–5 min		
Stirring after CpH addition	10 min		
$FeCl_2$ addition rate / stirring after	20 drops min^{-1}, 30 min		
Stirring after acid quench	15–20 min		
Vacuum drying	~30 min		

3. Colour checkpoints — two of these tell you the reaction is working

Stage	Expected	Observed / time	✓
KOH in DME, before CpH	white undissolved solid, colourless liquid	(pale)	
10 min after CpH	**deep burgundy / red-brown.** A pale pink means deprotonation is incomplete	pinkish brown?	
$FeCl_2 \cdot 4H_2O$ in DMSO	**pale green.** Yellow or orange means Fe^{2+} is already oxidising		
During the $FeCl_2$ addition	darkens to a dark orange-brown slurry		
On quenching into ice-acid	orange precipitate; some gas evolution	green? + bright orange + some bubbling	
Filtrate and washings	pale yellow-green, fading to colourless		
Crude ferrocene, dried	orange to orange-brown powder		

The two bold rows are the checks worth making *before* committing the next reagent. A burgundy solution says the cyclopentadienide formed; a pale pink says stir longer. A green iron solution says the Fe^{2+} is still intact; if it has gone yellow or orange, oxidation has already cost you yield and the low result is explained. Write what you see, not what is printed.

4. Crude product

Mass of empty vial	1.5428.	N_2 flow maintained throughout ☐✓	ok
Mass of vial + crude	1.6127	Gas evolution on quenching ☐	slight?
Mass of crude ferrocene	0.07???	Microscopic image taken ☒	a/r/orange

To be completed at the end of the session		
Student signature	GTA name	GTA signature Song Yan 10/9/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/10

WEEK 6 (Mon. / Thu.)

INORGANIC EXPERIMENT 2 — SESSION II

Purification and Characterization of Ferrocene

Venue	General & Synthetic Chemistry Lab, Block S5 Level-1	**Instrument / technique**	Stereomicroscope, MSB, IR, UV–Vis, ESI-MS, ^{1}H NMR
Time	10:00 a.m. – 4:00 p.m. (Mon. or Thu.)	**Source**	Manual v0724 + lecture notes

(a) Purification by sublimation

1. **Weigh both larger petri dish covers empty** on the analytical balance and record each.
2. Spread the crude ferrocene on the smaller petri dish base; cover with a weighed cover.
3. Raise the hotplate *slowly*; **100 °C for cycle 1**, and never past **150 °C**. Place the 400 mL beaker of ice water on the cover.
4. Take the cover off the heat, let it cool, recover the sublimed ferrocene. Repeat until nothing further sublimes.
5. Weigh each loaded cover; transfer the product to a clean transparent bag or vial.

WHAT TO EXPECT FROM EACH CYCLE — SEE APPENDIX G

Cycle 1 mostly moves **water**: the crude is still damp from last week's filtration, so expect a wet, pale cover and little orange. Raise the temperature from cycle 2 or 3 and the orange flakes appear. Non-volatile impurities stay on the *base*, which darkens as the cycles proceed — that residue is the evidence that purification happened.
Let the cover cool before handling, invert it over the bag and not the bench, and re-weigh the cover empty before reusing it. Only two are issued.

(b) Characterization — six measurements

1. **Stereomicroscopy.** 14×, one image, matriculation number typed on it. Compare against the Week 5 crude image; this pair is the qualitative evidence for purification.
2. **Magnetic susceptibility.** Grind **0.09–0.12 g** in the agate mortar. Both χ_v and χ_g, five photographs. *Read Appendix H before computing a moment; Appendix I shows why the answer must be zero.*
3. **UV–Vis.** ~~35–40 mg~~ 20–25 mg, weighed to four decimals, in methanol, **25 mL volumetric flask**. Record 300–700 nm; dilute until $A = 0.1$–1.0. Determine ε for each peak; the calculation must be handwritten on the spectra.
4. **IR.** Solid spectrum 500–4000 cm^{-1}, ATR or KBr. Record which. *Appendix C.*
5. **^{1}H NMR.** **~1 mg** into a tube containing **0.6 mL $CDCl_3$**. One sharp singlet near δ 4.15 for all ten protons is the result — the ten C–H are equivalent because both rings rotate freely and all five carbons of each bind iron equally.
6. **ESI-MS.** ~~1–2 mg~~ 5mg into a sample tube. *Appendix B.*

THE MSB PROCEDURE — FIVE PHOTOGRAPHS, ALL ASSESSED

Switch on and wait for a zero reading (−0.000E−5Z) [**photo 1**]. Weigh an empty tube, place it on the detector, read, press TARE, confirm zero. Remove it, introduce the ground sample, measure the packed length (**15.0–25.0 mm**) and mass (**0.0010–0.9999 g**). Press LENGTH and enter three digits [**photo 2, before the last digit**]. Press WEIGHT and enter [**photo 3, before the last digit**]. Press MAG SUS and record χ_v [**photo 4**] and χ_g [**photo 5**]. The display advances the instant the last digit is entered.

Before you leave: ☐ Purified ferrocene labelled ☐ Microscope image ☐ Five MSB photos ☐ IR and UV–Vis spectra ☐ ESI-MS and NMR submitted ☐ Datasheet signed

original sample: microscopy ✓ → one more for extra sample → 1.4447 g initially (with vial)
^{1}H NMR sample submission ✓
ESI-MS sample submission
ATR-IR
UV-Vis.

1.1083
− 1.0989
(extra 20 mg)
→ new (extra sample

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (1 of 2)

Session II: Sublimation Record

Full name (as on student ID)	Matriculation no.	Date
KZH DONGHUIGH.	A0266 094K	2026/09/17.
Lab session ☐ Mon ☑ Thu	Fumehood no. 137.	S/N no. assigned by lab officer

1. Starting material

petri s : 58.9177 g

Mass of crude ferrocene from Week 5		Hotplate setting, cycle 1	
Cover A, empty (g) Left	48.6311	Cover B, empty (g) right.	48.7280

2. Sublimation cycles

extra 45.7688

extra 48.6315

Cycle	Set / °C	Cover	Cover + Fc (g)	Fc (g)	Appearance of cover / base
1	~100°C	A / B	48.6714	0.0403	} 0.0526 g total
2	~140°.	A / B	45.7843	0.0123	
3		A / B	45.7792	0.1104	
4		A / B	48.6948.	0.0563	
			Total purified ferrocene (g)		58.9315 − 58.9177 = 0.0138

new (extra)

extra: 0.1805 g

3. Colour checkpoints

Stage	Expected	Observed / time	✓
Crude, before heating	orange to orange-brown powder		
Cover after cycle 1	**wet and pale** — mostly condensed water, little orange		
Cover after cycles 2–3	orange flakes or plates		
Petri dish **base**, later cycles	darkens; non-volatile residue left behind		
Purified product	bright orange crystalline flakes		
After grinding for the MSB	light orange free-flowing powder		
Methanol solution for UV	clear yellow-orange		

Expected is a checkpoint, not an answer. A pale cover after cycle 1 is the normal result, not a failure; record what you see and the time.

To be completed at the end of the session		
Student signature	GTA name	GTA signature Sang Jun 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/17 page 1.

1-vial 0.0526 g

Submit extra sample vial two

extra : 0.3021 g.

after sublimation/purification :

empty vial : 1.0961 g

1.3982

1.1845

0.0137

0.0945

DATASHEET — INORGANIC EXPERIMENT 2 — WEEK 6 (2 of 2)

Session II: Characterization Record

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

4. Magnetic susceptibility — Sherwood MSB AUTO

Zero reading		Packed length L (mm)	17 mm
Empty tube mass (g)		Sample mass m (g)	0.0867
Volume susceptibility χ_v	−0.005	Mass susceptibility χ_g	0.000
Room temperature (K)		Sign of χ_g ☐ − ☑ 0 ☐ +	

Photographs: ☐ zero ☐ before last digit of length ☐ before last digit of weight ☐ χ_v ☐ χ_g **A reading of exactly zero is a resolution limit, not a moment — Appendix H.**

5. UV–Vis dilute until $A = 0.1 \sim 1.0$. → look for ε.

Mass weighed (g, four decimals)		Path length (cm)	1.00
Flask volume (mL) / solvent	25, MeOH	Concentration c (M)	
Dilution factor, if any		λ_1 (nm) / A_1	
Expected λ: ~440 and ~325 nm		λ_2 (nm) / A_2	

6. IR, NMR, ESI-MS and imaging

IR method ☑ ATR ☐ KBr		Mass for ^{1}H NMR (~1 mg)	
Microscope image of purified ☐		Mass for ESI-MS (1–2 mg)	
S/N no. for NMR		S/N no. for ESI-MS	

7. Yield and observations

Purified mass / crude mass, and the % recovered	Crude vs purified under the microscope — size, habit, transparency

To be completed at the end of the session		
Student signature	GTA name	GTA signature
		Cheng Thu 17/09/26

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

2026/09/17 page 12

photo - control unit and magnetic susceptibility serial #.

17 mm.

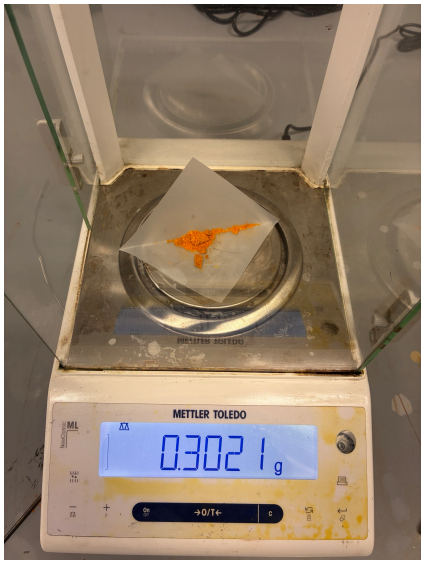
1. zero ②
2. ~~control~~ unit ⑦ length
3. weight. ⑧
4. control ⑨
5. balance.

total 5 photos.

0.8988
0.8223
0.0765 g.

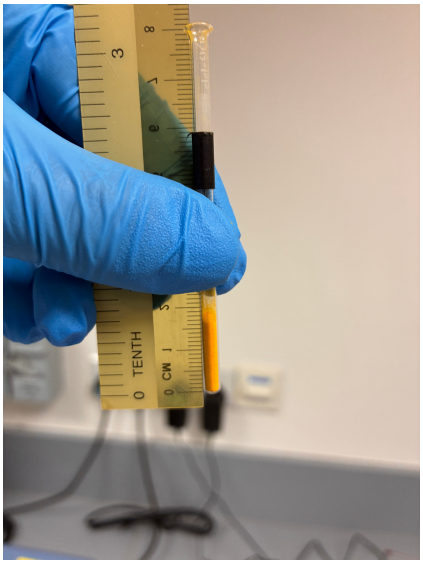
0.9090
− 0.8264
0.0826.

Appendix E. Weighing and magnetic-balance photographs



(a) second portion of crude ferrocene on tared weighing paper, 0.3021 g (IMG_8546)

(b) filled MSB tube on the balance, 0.9090 g (IMG_8571)



(c) packed sample beside the ruler, 17 mm (IMG_8572)

Figure 9: Weighing and packed-length photographs (original files).

Figure 10: Enlarged display crops: volume-mode result, mass-mode result, the zero-reading frame (IMG_8576.JPG; the digits are not resolvable even after contrast enhancement) and the handwritten mass subtraction on the datasheet (both lines use 0.8223 g as the empty-tube mass).